7 Week 6: Foundations for Causal Inference in Observational Studies

7.1 Motivation: From Randomized Experiments to Observational Studies

In Weeks 2–4, our causal conclusions rested on a simple but powerful fact: treatment assignment was randomized. Under randomization, treated and control units are comparable in expectation on all covariates, whether measured or unmeasured. Probability statements about estimators and test statistics arise from the known randomization distribution.

In many scientific and policy settings, however, randomization is infeasible, unethical, or impossible. Treatments are assigned through human decisions, institutional rules, or self-selection. These processes typically depend on unit characteristics that also affect outcomes. As a result, treated and control groups in observational data are generally not comparable without adjustment.

Our goal in Weeks 6–7 is to understand how, under explicit assumptions, observational studies can be analyzed in a way that approximates the logic of randomized experiments. We will:

- formalize the assumptions needed for identification,
- introduce the propensity score and its balancing property,
- study matching and weighting as design tools,
- derive inverse probability weighting (IPW) and augmented IPW (AIPW),
- emphasize design-based thinking in observational studies.

Throughout, we restrict attention to a binary treatment $Z \in \{0, 1\}$.

7.2 Potential Outcomes and the Source of Bias

For each unit $i = 1, \ldots, N$, define potential outcomes

$$Y_i(1), \qquad Y_i(0),$$

representing the outcome that would be observed under treatment ($Z_i = 1$) and control ($Z_i = 0$), respectively. The observed outcome is

$$Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

Our target causal parameter is the average treatment effect (ATE)

$$\tau = \mathbb{E}[Y(1) - Y(0)].$$

In observational data, a naive comparison of observed outcomes is

$$\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0].$$

We can decompose this difference as follows:

$$\begin{aligned}
\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0] &= \mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0] \\
&= \big(\mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 1]\big) + \big(\mathbb{E}[Y(0) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0]\big) \\
&= \tau + \underbrace{\big(\mathbb{E}[Y(0) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0]\big)}_{\text{selection bias}}.
\end{aligned}$$

The second term is *selection bias*. It arises because treated and control units differ systematically even before treatment is applied. Eliminating or controlling this term is the central challenge of causal inference in observational studies.

7.3 SUTVA: The Stable Unit Treatment Value Assumption

The potential outcomes framework requires that the treatment assigned to one unit does not ambiguously affect the potential outcomes of another unit, and that the treatment itself is well-defined. These requirements are formalized in the *Stable Unit Treatment Value Assumption* (SUTVA). SUTVA has two components.

1. No interference between units. For each unit i, the potential outcomes depend only on its own treatment assignment:

$$Y_i(z_1, \ldots, z_N) = Y_i(z_i).$$

This assumption rules out spillovers, contagion, peer effects, and any situation in which the treatment received by one unit affects the outcome of another. Examples of interference include:

- vaccination campaigns where one person's vaccination affects others' infection risk,
- classroom interventions where treated students influence untreated peers,
- network settings where behavior spreads through social ties.

When interference is present, the potential outcomes must be indexed by the entire assignment vector, and the standard ATE is no longer well-defined without additional structure.

2. No hidden versions of treatment. SUTVA also requires that the treatment $Z = 1$ and control $Z = 0$ correspond to unique, well-defined interventions. Formally, there must not exist multiple "versions" of treatment that produce different potential outcomes. Examples of violations include:

- job training programs with heterogeneous curricula or instructors,
- medical treatments with varying dosages or delivery methods,
- educational interventions implemented differently across schools.

If multiple versions exist, the potential outcomes $Y_i(1)$ and $Y_i(0)$ are ambiguous unless the versions are explicitly modeled.

Interpretation. SUTVA ensures that potential outcomes are meaningful objects. Without SUTVA, the notation $Y_i(1)$ and $Y_i(0)$ does not uniquely describe what would happen under treatment or control. Randomized experiments often satisfy SUTVA by design, but observational studies require careful argumentation to justify it. In practice, researchers must:

- define the treatment precisely,
- consider whether units may affect one another,
- assess whether multiple versions of treatment exist,
- justify why interference or hidden versions are negligible or irrelevant.

SUTVA is conceptually prior to ignorability and overlap: even if treatment assignment is ignorable and overlap holds, causal effects are not identifiable unless potential outcomes are well-defined.

7.4 Ignorability of Treatment Assignment

We now formalize the assumption that, conditional on observed covariates, treatment assignment behaves "as if" randomized.

Definition 1 (Ignorability / Unconfoundedness)**.** *Treatment assignment is ignorable given covariates X if*

$$(Y(1), Y(0)) \perp Z \mid X.$$

Ignorability asserts that, after conditioning on X, there are no unmeasured confounders that jointly affect treatment assignment and potential outcomes.

Interpretation. Ignorability does not claim that treatment is random. Rather, it asserts that all variables that jointly affect treatment assignment and potential outcomes have been measured and included in X. This is a strong assumption and cannot be verified from the data alone.

Here X are our observed confounders. The ignorability assumption states that there are **no unobserved confounders**, the covariates X includes all observed confounders.

Design based perspective. Ignorability is a *design* assumption, not a statistical property of the data. Its plausibility must be justified by:

- subject-matter knowledge,
- understanding of how treatment decisions are made,
- careful selection of covariates,
- sensitivity analyses for unmeasured confounding.

7.5 Overlap and Common Support

Ignorability alone is not sufficient for identification. We also require that, for each covariate profile, both treatment and control units are possible.

Definition 2 (Overlap / Common Support)**.** *The overlap condition requires*

$$0 < \Pr(Z = 1 \mid X = x) < 1 \quad \textit{for all } x \textit{ with positive density.}$$

Overlap ensures that, for every covariate profile x, there exist both treated and control units in the population. Without overlap, identification of τ requires extrapolation beyond the support of the data.

Practical implications. Overlap is a design issue, not a modeling issue. When overlap fails:

- no amount of modeling can recover the causal effect without strong assumptions,
- estimators become unstable (e.g., extreme weights in IPW),
- trimming or restricting to regions of common support is often necessary.

Diagnostics. Common diagnostics for overlap include:

- histograms or density plots of estimated propensity scores by treatment group,
- minimum and maximum estimated propensity scores in each group,
- overlap plots (e.g., boxplots or violin plots of $e(X)$ by Z),
- effective sample size under weighting.

7.6 Identification Under Ignorability

Assume SUTVA, ignorability and overlap. We show that the ATE is identified from the joint distribution of (Y, Z, X).

First, note that

$$\mathbb{E}[Y(1)] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid X]\big].$$

This is a direct result of the tower property of conditional expectation.

By ignorability,

$$\mathbb{E}[Y(1) \mid X = x] = \mathbb{E}[Y(1) \mid Z = 1, X = x].$$

whenever $\Pr(Z = 1 \mid X = x) > 0$.

By consistency ($Y = Y(1)$ when $Z = 1$),

$$\mathbb{E}[Y(1) \mid Z = 1, X = x] = \mathbb{E}[Y \mid Z = 1, X = x].$$

Thus, invoking the overlap assumption,

$$\mathbb{E}[Y(1)] = \mathbb{E}\big[\mathbb{E}[Y \mid Z = 1, X]\big].$$

Analogously,

$$\mathbb{E}[Y(0)] = \mathbb{E}\big[\mathbb{E}[Y \mid Z = 0, X]\big].$$

Therefore,

$$\boxed{\tau = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}\big[\mathbb{E}[Y \mid Z = 1, X] - \mathbb{E}[Y \mid Z = 0, X]\big].}$$

This expression shows that causal effects can be recovered by comparing treated and control units with the same covariates X.

7.7 The Propensity Score: Definition and Balancing Property

We now introduce the central object for covariate adjustment in observational studies.

Definition 3 (Propensity Score)**.** *The propensity score is*

$$\boxed{e(X) = \Pr(Z = 1 \mid X).}$$

The propensity score summarizes, in a single number, how likely a unit is to receive treatment based on its covariates.

Theorem 5 (Rosenbaum–Rubin Balancing Property)**.** *If treatment assignment is ignorable given X, then*

$$\mathrm{X} \perp Z \mid e(X).$$

Moreover, $e(X)$ is a balancing score*: conditional on $e(X)$, the distribution of X is the same in treated and control groups.*

Interpretation. Within strata of the propensity score, treated and control units have the same distribution of covariates. Thus, conditioning on $e(X)$ balances X in the same sense that randomization balances covariates in expectation. Thus, it is enough to compare treated and control units that are similar just in their propensity scores.

Proof (sketch; you can skip). We show that $Z \perp X \mid e(X)$, which implies that conditioning on $e(X)$ balances X.

Fix x such that $e(x) = p$. Then

$$\Pr(X = x \mid Z = 1, e(X) = p) = \frac{\Pr(Z = 1 \mid X = x)\Pr(X = x)}{\Pr(Z = 1 \mid e(X) = p)}.$$

But $\Pr(Z = 1 \mid X = x) = e(x) = p$, and $\Pr(Z = 1 \mid e(X) = p) = p$ as well. Thus,

$$\Pr(X = x \mid Z = 1, e(X) = p) = \frac{p\Pr(X = x)}{p\Pr(X \in \{x : e(x) = p\})} = \frac{\Pr(X = x)}{\Pr(X \in \{x : e(x) = p\})}.$$

The same expression holds for $Z = 0$, so

$$\Pr(X = x \mid Z = 1, e(X) = p) = \Pr(X = x \mid Z = 0, e(X) = p).$$

Hence $X \perp Z \mid e(X)$, and therefore $(Y(1), Y(0)) \perp Z \mid e(X)$ under ignorability.

□

7.8 Geometry of Balancing Scores

A *balancing score* is any function $b(X)$ such that

$$Z \perp X \mid b(X).$$

The propensity score $e(X)$ is the *coarsest* balancing score: if $b(X)$ is any balancing score, then $e(X)$ is a function of $b(X)$.

The geometry is:

- X is high-dimensional (e.g., age, education, income, comorbidities),
- $e(X)$ is one-dimensional,
- conditioning on $e(X)$ balances all functions of X.

This reduction from high-dimensional X to a scalar $e(X)$ is the key to practical causal inference in observational studies.

7.9 Estimating the Propensity Score

In practice, $e(X)$ is unknown and must be estimated. Common approaches include:

- logistic regression: $\Pr(Z = 1 \mid X) = \text{logit}^{-1}(\alpha + \beta^\top X)$,
- probit models,
- flexible machine learning methods (e.g., random forests, boosting, neural networks).

Design perspective. The goal of propensity score estimation is not prediction of Z per se, but construction of weights or matched sets that achieve good covariate balance. A model with excellent predictive performance (e.g., high AUC) may still yield poor balance if it overfits or extrapolates.

Model checking. After estimating $e(X)$, we:

- examine overlap in estimated scores,
- assess covariate balance after matching or weighting,
- refine the model if balance is inadequate.

8 Week 7: Methods for Adjustment — Matching, Weighting, and IPW